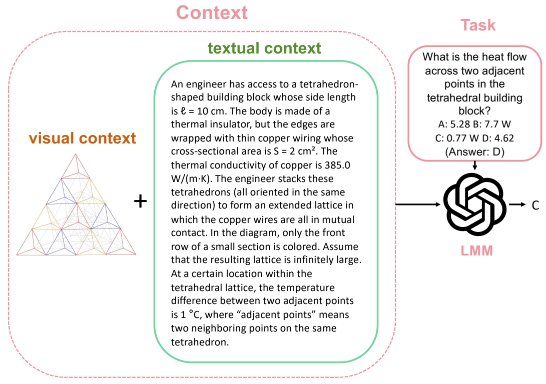
Context
visual context
textual context
An engineer has access to a tetrahedron-shaped building block whose side length is ℓ = 10 cm. The body is made of a thermal insulator, but the edges are wrapped with thin copper wiring whose cross-sectional area is S = 2 cm². The thermal conductivity of copper is 385.0 W/(m·K). The engineer stacks these tetrahedrons (all oriented in the same direction) to form an extended lattice in which the copper wires are all in mutual contact. In the diagram, only the front row of a small section is colored. Assume that the resulting lattice is infinitely large. At a certain location within the tetrahedral lattice, the temperature difference between two adjacent points is 1 °C, where "adjacent points" means two neighboring points on the same tetrahedron.
+
Task
What is the heat flow across two adjacent points in the tetrahedral building block?
A: 5.28 B: 7.7 W
C: 0.77 W D: 4.62
(Answer: D)
LMM
C